END BEHAVIOR

Horizontal asymptotes describe the end behavior of a rational function. Which one appears depends only on how the degree of the numerator compares with the degree of the denominator.

RATIONAL FUNCTION

$$f(x) = \frac{P(x)}{Q(x)}$$

split by $\deg P$ compared with $\deg Q$

=

CASE 1

$\deg P = \deg Q$

Ratio of leading coefficients

$$y = \frac{a_n}{b_m}$$

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CASE 2

$\deg P < \deg Q$

The x-axis

$$y = 0$$

E.G. $f(x) = \frac{1}{x}$, horizontal asymptote $y = 0$

>

CASE 3

$\deg P > \deg Q$

No horizontal asymptote

an oblique asymptote appears when the degree difference is exactly 1

TAKEAWAY

Vertical asymptotes and holes come from the zeros of Q; horizontal asymptotes come from comparing degrees. The two questions are separate.